# BS Computer Engineering Suggested Four Year Plan
# 2019 – 2020 Catalog

**A grade of C- or better is required in each class.**

****A grade of C or better is required in CS 1336, CE 1337, CE 2305, CS 3377, CS 2337, and CE 2336.**

CS 1336 is required of students with no prior programming experience
**Not For Credit in ECS**
**C or better needed**

****CS 1336**
**CS 1136**

## Freshman

| Description | Course No |
|---|---|
| Introduction to Comp Eng | CE 1100 |
| Introduction to Engr'g. & CS | ECS 1100 |
| Calculus I | Math 2413 or 2417 |
| Computer Science I | CE 1337 |
| 6 hours CORE: Choices below | |
| Freshman Seminar | UNIV 1010 |

ECS 1100 → UNIV 1010

CE 1100

Math 2413 → Math 2417

**CE 1337

CORE

CORE

## Freshman

| Description | Course No |
|---|---|
| Intro to Elec. Engineering | CE 1202 |
| Calculus II | Math 2414 or 2419 |
| Introduction to Digital System | CE 2310 |
| Applied Linear Alegebra | ENGR 2300 |
| Mechanics and Heat | Phys 2325 |
| Mechanics and Heat Lab | Phys 2125 |
| Calculus of Several Variables (Summer) | MATH 2415 |

CE 1202

Phys 2325
Phys 2125

Math 2414 → Math 2419

ENGR 2300

CE 2310

Math 2415

Calculus Complete

## Sophomore

| Description | Course No |
|---|---|
| Differential Equations w/Applications | Math 2420 |
| Computer Science II | CE 2336 or 2337 |
| Probability & Stats in CS and SE | ENGR 3341 |
| Electricity and Magnetism | Phys 2326 |
| Electricity and Magnetism Lab | Phys 2126 |
| Discrete Mathematics I | CE 2305 |

Phys 2326
Phys 2126

Math 2420 → Recommended → ENGR 3341

AP score at least 4

**CE 2336 **OR** **CS 2337

Math 2312 (C or better) or 80% on ALEKS

**CE 2305

## Sophomore

| Description | Course No |
|---|---|
| Electrical Network Analysis | CE 3301 |
| Fundamentals I | CE 3201 |
| Data Structures & Intro to Algorithmic Analysis | CE 3345 |
| Advanced Engineering Math | ENGR 3300 |
| Digital Circuits | CE 3320 |
| 3 hours Core: Choices below | |

CE 3201

RHET 1302

CE 3320

CE 3301

CE 3345

ENGR 2300

ENGR 3300

CE 3320

CORE

## Junior

| Description | Course No |
|---|---|
| Electronic Devices | CE 3310 |
| Software Engineering | CE 3354 |
| Prof & Tech Communication | ECS 3390 |
| 6 hours CORE: Choices below | |

**CE 2305 & **CE 2336

Junior & Rhet 1302

ECS 3390

CE 3354

CE 3310

CORE

CORE

## Junior

| Description | Course No |
|---|---|
| Fundamentals II | CS 3202 |
| Electronic Circuits | CE 3311 |
| Computer Networkds | CE 4390 |
| Signals ans Systems | CE 3302 |
| 6 hours CORE: Choices below | |

CE 3202

CE 3311

CE 4390

CE 3302

CORE

CORE

## Senior

| Description | Course No |
|---|---|
| Senior Design Project I | CE 4388 |
| Computer Architecture | CE 4304 |
| CE Elective Lab | CE 420X |
| Social Issues & Ethics in Sci/ Tech | ECS 2361 |
| 3 hours Free Elective | See Advisor |

CE 420X

Free Elective

CE 4388

CE 4304

ECS 2361 ①

C & UNIX Knowledge

**CE 2305 & **CE 2336

## Senior

| Description | Course No |
|---|---|
| Senior Design Project II | CE 4389 |
| Operating Systems Concepts | CE 4348 |
| Embedded Microprocessor Sys. | CE 4370 |
| Org. Of Programming Lang. | CE 4337 |
| 3 hours Free Elective | See Advisor |
| Core Curriculum Assessment | UNIV 2020 |

UNIV 2020

Free Elective

CE 4389

CE 4370 (Spring Only)

CE 4348

CE 4337

### Core Choices

| Rhet 1302 | 3 hours From: | 3 hours From: | 6 hours From: |
|---|---|---|---|
| Govt 2305<br>Govt 2306 | Ahst 1303, 1304, 2331<br>Arts1301<br>Danc 1310<br>Film 2332<br>Musi 1306<br>Thea 1310 | Ams 2300<br>Ams 2341<br>Huma 1301<br>Lit 2331<br>Phil 1301<br>Phil 2316, 2317 | Hist 1301<br>Hist 1302<br>Hist 2301<br>Hist 2330<br>Hist 2332 |

**① Pre-Req. of PHYS 2325, PHYS 2326 or any other 030 science core**

**Courses may be taken in any order, as long as prerequisites are met.**

LEGEND

Prereq Critical Path

Pre-requisite

Pre- or Co-requisite